When a Warning Advertises: The Boomerang Effect in Anti-Drug Campaigns and the Case of Concentrated 7-Hydroxymitragynine

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Author note. The author trained in family medicine and previously ran a direct primary care clinic as its chief executive. He is not currently licensed to practise medicine. This article is a scholarly commentary on public health communication and drug policy. It makes no clinical or medication recommendations, and nothing in it is medical advice for any individual. Treatment decisions belong to patients and their own clinicians. The author also writes from lived experience of substance use and recovery; that perspective is identified where it appears and is presented alongside the published evidence, not in place of it.

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Abstract

Public service announcements and prohibition announcements share an assumption: that naming a danger reduces it. A body of evidence from health communication, prevention science and drug policy says the assumption fails often enough to plan for. Psychological reactance, the motive to restore a threatened freedom, has been confirmed in meta-analysis as a response to controlling health messages. The United States National Youth Anti-Drug Media Campaign was associated in its own evaluation with weaker intentions to avoid marijuana among more exposed young people, a Cochrane review of mass media prevention found pooled effects on drug use that were null with iatrogenic effects in individual campaigns, and meta-analyses of the DARE programme found effect sizes on drug use close to zero. Tobacco industry campaigns aimed at parents were associated with stronger intentions to smoke among older students, while the autonomy-respecting "truth" campaign accounted for a measurable share of a national decline in youth smoking. The counterevidence is part of the record too: a large meta-analysis of fear appeals found positive average effects and no identified backfire conditions, and interactive school prevention outperformed lecture-based models, which together locate the risk in message design and context rather than in communication itself. This commentary applies that record to the current United States response to concentrated 7-hydroxymitragynine (7-OH), a potent opioid alkaloid sold in tablets, gummies and shots, which federal regulators moved against in 2025. No direct evidence yet shows that publicity about 7-OH has increased curiosity or use, and this article claims none. It argues, from adjacent evidence including the United Kingdom mephedrone episode of 2009 to 2010, where news coverage functioned in part as advertising, that publicity around a little-known drug carries a measurable risk of raising awareness and purchase-directed curiosity, and that campaign design, individualised care planning and non-stigmatising disease framing reduce that risk. It also questions the permanent identity of "addict" that recovery culture often assigns, contrasting it with the way other chronic diseases are framed, and sets out design principles for campaigns and care that warn without advertising.

Keywords: psychological reactance; boomerang effect; health communication; kratom; 7-hydroxymitragynine; mephedrone; harm reduction; stigma; addiction

Introduction

Every anti-drug campaign begins with an act of publicity. Before a message can warn a single person away from a substance, it has to tell that person the substance exists, what it is called, and often what it does. For well-known drugs that cost is trivial, because the audience already knows. For a little-known drug, sold under a name most people have never heard, the cost is the whole campaign. The warning is the introduction.

That tension sits at the centre of the current debate over kratom and its concentrated alkaloid 7-hydroxymitragynine, shortened to 7-OH. Through 2025, United States regulators issued warning letters, seized products and recommended federal scheduling of concentrated 7-OH, and each step required press releases, news coverage and public explanation of a product most Americans had never encountered. The regulatory case may be sound. This commentary asks a separate question that the regulatory case does not answer: what does the publicity itself do, and who does it reach?

The question is not new, and it has a name. Communication researchers call it the boomerang effect: a persuasive message that produces the opposite of its intended effect in part of its audience [1,2]. This article reviews what is known about boomerang effects in drug and tobacco prevention, sets out the current facts on kratom and 7-OH as regulators themselves state them, and then examines a case the author knows from the inside, the British mephedrone wave of 2009 to 2010, where the published record shows news coverage acting, in part, as a sales channel. The second half of the article turns from campaigns to care. It argues that addiction should be treated as a disease, that the disease model is let down by the permanent identity language wrapped around it, and that care built around the individual, with goals the person chooses, psychological support, harm reduction and a wide circle of stakeholders, fits the evidence better than either fear-driven messaging or open-ended labels.

One boundary governs everything that follows. This is a commentary on communication and policy, written by an author who is not currently licensed to practise. It recommends no medication and no treatment for any individual reader. Where treatment research is discussed, it is discussed as published evidence about systems and programmes.

Kratom and 7-OH: what regulators say, and what the data can and cannot show

Kratom (Mitragyna speciosa) is a Southeast Asian tree whose leaves have a long history of traditional use and are now sold in the United States as powders, capsules, teas and extracts. The United States Food and Drug Administration (FDA) Center for Drug Evaluation and Research assessment of 7-OH, prepared by Reissig and colleagues in 2025, records that 7-OH makes up roughly 0.01 to 0.04 percent of dry leaf weight, that it binds the mu-opioid receptor with high affinity, and that concentrated products on the market are most plausibly made by chemically converting the more abundant alkaloid mitragynine, since extracting that share from leaf directly makes little economic sense [3]. Laboratory work by Kruegel and colleagues in 2016 characterised mitragynine and 7-OH as partial agonists at the human mu-opioid receptor [4]. Case reports in the addiction medicine literature describe dependence and opioid-like withdrawal in people using high-dose 7-OH tablets bought in smoke shops [5].

Two cautions belong in any honest account. First, potency comparisons with morphine that circulate in press coverage come from animal and laboratory assays, vary widely between studies, and have no

human trial data behind them; the FDA and Department of Health and Human Services review puts the figure at roughly two to nine times morphine depending on the study [3]. This article prints no single potency figure as a human fact. Second, the best population data cannot separate the leaf from the concentrates. Towers and colleagues, reporting poison centre data in MMWR in 2026, counted kratom-related reports rising from 258 in 2015 to 3,434 in 2025, with 233 deaths over the period, 79 percent of them involving multiple substances [6]. The authors state plainly that their data cannot distinguish traditional leaf products from concentrated or semisynthetic ones. The rise in reports is real. Its attribution to 7-OH alone is not established by that dataset. Survey data put lifetime kratom use at 1.9 percent of Americans aged 12 and over in 2024, about 5.4 million people [7].

The regulatory actions of 2025 were aimed at the concentrates. On 15 July 2025 the FDA sent warning letters to seven firms marketing 7-OH tablets, gummies, drink mixes and shots, sold online and in gas stations, smoke shops and corner stores [8]. On 29 July 2025 the agency recommended scheduling concentrated 7-OH under the Controlled Substances Act, and Commissioner Makary drew the line in public: "We're not targeting the kratom leaf or ground-up kratom. We are targeting a concentrated synthetic byproduct that is an opioid" [9]. In December 2025 federal marshals seized roughly 73,000 units of 7-OH products from Missouri warehouses [10]. In July 2026 the Drug Enforcement Administration published a notice of intent to begin temporary Schedule I scheduling of 7-OH above defined thresholds, a notice of intent being a step in a process rather than final scheduling [11]. States moved in parallel: six states and the District of Columbia had banned kratom in all forms by August 2025, while others, including Florida, Ohio, Colorado and Oklahoma, targeted concentrates, set age limits or capped 7-OH content [12]. Expert opinion is split in a way that matters for this article: Henningfield and colleagues, who have published eight-factor analyses of kratom, support Schedule I control of concentrated and synthetic 7-OH while preserving lawful access to natural leaf, and other specialists argue in the press for lower scheduling or a regulatory framework that keeps research open [13].

None of those facts answer the communication question. A decision to schedule a substance and a decision about how loudly, in what words, and with what product detail to announce it are two different decisions. The first belongs to regulators working from toxicology. The second is a health communication intervention delivered to the whole population, including people who had never heard of the drug, and it should be held to the standards applied to any other population intervention: evidence of benefit, monitoring for harm.

The boomerang effect: mechanism and evidence

Psychological reactance was described by Brehm in 1966: when a person perceives a threat to a behavioural freedom, a motivational state arises aimed at restoring that freedom, and its strength grows

with the importance of the freedom and the size of the threat [14]. A teenager told, in effect, "you must not" has something to push against, and pushing can mean doing the prohibited thing. Rains tested the mechanism in a 2013 meta-analysis of 20 studies and 4,942 participants, and found that anger and counterarguing rise together as indicators of reactance, with anger the stronger of the two [15]. Reviews of the health communication literature since then have tied reactance to controlling language, threat appeals and perceived manipulative intent [16]. Byrne and Hart, reviewing unintended effects in 2009, described two routes to a boomerang: audiences attend to the harmful or unintended elements of a message at the expense of the intended ones, or they process the message as intended and resist complying with it [1]. A related literature on media ratings calls the same pull the forbidden fruit effect: warnings and restrictions can make restricted material more attractive to adolescents, an effect Bushman and Cantor documented for violent media in 2003 and which transfers by mechanism, if not by direct trial, to prohibited substances [17].

At campaign scale, the flagship case is the United States National Youth Anti-Drug Media Campaign of 1999 to 2004. Hornik and colleagues evaluated it in the American Journal of Public Health in 2008 and found no favourable effect overall, and an unfavourable pattern underneath: greater exposure predicted weaker intentions to avoid marijuana, and exposure in one survey round predicted marijuana initiation in the next [18]. The authors were careful with their own finding, calling the pro-marijuana association "tentative but not definitive," and this article repeats their hedge, because the design was observational. A Cochrane review by Ferri and colleagues in 2013 pooled the randomised evidence on mass media prevention in young people and found no effect on drug use, while noting clear iatrogenic effects in individual campaigns among the non-pooled studies [19]. The pattern across the literature is consistent in shape even where it is modest in size: mass anti-drug messaging tends to do nothing on average, and in identifiable conditions it does harm.

DARE: thirty years of null results

The DARE programme (Drug Abuse Resistance Education) put police officers in classrooms across America through the 1980s and 1990s and became the most widely delivered school drug prevention programme in the country. Its evaluation record is unusually complete, and unusually flat. Ennett and colleagues meta-analysed eight rigorous evaluations in 1994 and found effect sizes on drug use between 0.00 and 0.11, with a weighted mean of 0.06; the programme's largest effect was on knowledge, at 0.42 [20]. West and O'Neal updated the meta-analysis in 2004 and found a weighted effect of r = 0.011 (d = 0.023) on drug use, nonsignificant, a result they summarised as virtually no effect [21]. Lynam and colleagues followed 1,002 children randomised in sixth grade through to age 20 and reported in 1999 that there were few differences in drug use, attitudes or self-esteem, and that "in no case did the DARE group have a more successful outcome" [22]. Rosenbaum and Hanson's randomised six-year study found no long-term effect overall, with small but significant increases in drug use in a suburban subgroup [23]. A

2003 review by the US Government Accountability Office reached the same destination by a different road: six long-term evaluations showed no statistically significant differences in drug use [24]. These findings describe the original DARE curriculum, and later curricula should be judged on their own data. Ennett's team added a comparison that points forward rather than back: interactive prevention programmes, built on discussion and skills practice instead of lectures, outperformed DARE in the same meta-analysis [20]. School-based prevention is not a failed idea. One delivery model failed, at scale, for decades, while a better-supported model sat beside it in the literature. The lesson is about the original model's ingredients: an authority figure, a classroom of adolescents, information about drugs many of them had not yet encountered, and a single message of refusal. Delivered at national scale for years, it taught effectively and prevented almost nothing.

Smoking: the campaign that recruited, and the campaign that worked

Tobacco offers the cleanest comparison, because two campaign styles ran at the same time in the same country. Philip Morris and other manufacturers funded advertisements aimed at parents, urging them to talk to their children about smoking. Wakefield and colleagues analysed their association with youth outcomes in 2006 and found that among students in grades 10 and 12, each additional viewing of the parent-targeted advertisements was associated with lower perceived harm of smoking, stronger approval of smoking, stronger intentions to smoke (odds ratio 1.12) and higher past-30-day smoking (odds ratio 1.12) [25]. Farrelly and colleagues found in 2002 that exposure to the Philip Morris "Think. Don't Smoke" youth campaign was not associated with anti-tobacco attitudes, and exposed young people were more likely to be open to smoking [26]. One randomised experiment complicates the story and belongs in it: Henriksen and colleagues reported in 2006 that industry advertisements did not raise intentions to smoke or curiosity overall in their experimental sample, although adolescents high in reactance had greater smoking intentions regardless of exposure [27]. The observational and experimental findings can both be true if the harm concentrates in the high-reactance segment of a real-world audience. That is, in fact, the segment prevention most needs to reach.

Against that background, the "truth" campaign reads as a controlled demonstration of how to handle reactance rather than provoke it. Farrelly and colleagues evaluated it in 2005: youth smoking prevalence fell from 25.3 percent to 18.0 percent between 1999 and 2002, and the campaign accounted for an estimated 22 percent of that decline [28]. Truth did not lecture adolescents about their behaviour. It aimed their distrust at the tobacco industry, documented its marketing practices, and let the audience draw its own conclusion. The reactance that other campaigns triggered against the messenger was redirected toward the seller. Grandpre and colleagues supplied the mechanism in 2003: explicit anti-smoking messages were associated with stronger future smoking intentions than implicit ones among tenth graders only, with no difference in younger children, which fits a reactance account in which autonomy concerns grow through adolescence [29].

One more finding keeps this literature honest. Palmgreen and colleagues, working on the SENTAR model, built television advertisements with high message sensation value and aimed them at high-sensation-seeking adolescents, the group at highest risk and the group most likely to react against a lecture. Their campaigns reversed upward trends in 30-day marijuana use among high sensation seekers [30]. A message can hold the attention of the exact audience most prone to boomerang, if it is built for that audience rather than for the adults funding it.

The strongest counterweight to the boomerang literature also belongs here, because an honest version of this argument has to survive it. Tannenbaum and colleagues meta-analysed the fear appeal literature in Psychological Bulletin in 2015: 127 articles, 248 independent samples, 27,372 participants. Fear appeals worked, on average, with a composite effect of d = 0.29 on attitudes, intentions and behaviour, and the authors reported no identified circumstances in which fear appeals backfired [59]. That finding deserves weight, and it has been disputed on method. Kok and colleagues replied in 2018 that the trials Tannenbaum pooled rarely tested the condition theory actually predicts trouble for, high threat paired with low confidence that the recommended action works, and that in the few experiments that did test it, behaviour moved, if anything, in the unhealthy direction [60]. The two results can be reconciled, and the reconciliation is the useful part. Fear with a credible way out tends to work. Fear without one, aimed at people who prize the threatened freedom, is where the backfire risk concentrates. Drug prohibition announcements sit closer to the second condition than the first for one specific reason: for a drug the audience has never heard of, the announcement carries awareness of the product before it carries anything else, and it offers most listeners no action to take beyond curiosity or avoidance of a thing they were not seeking in the first place. Smoking warnings never had that problem. Everyone already knew what a cigarette was.

The conclusion the evidence supports is narrow and useful: backfire is a design failure with known causes, among them controlling language, authority delivery, product salience without a change in norms, and a mismatch between message style and audience. It is not an argument against public communication. It is an argument for treating communication like any other intervention, designed against evidence and tested for harm before it is scaled.

Lived experience: mephedrone, and the news as a route in

This section is written in the first person. It records the author's experience, and the published evidence that sits around it.

I know the mechanism in the studies above from my own life. I am in recovery. During the British "legal high" years I was living in the United Kingdom, and the route by which grey-area chemicals reached me was the news. I read about mephedrone in newspaper coverage of a danger I had supposedly never heard

of, coverage that explained what it was, that it was legal, what it cost and where it was sold, in the course of warning people off it. The warning worked on me as an introduction. I experimented with substances I would not have known to ask for if the coverage had never run.

I do not offer that as evidence by itself. A single recollection is the weakest form of data there is, and my memory of those years has the obvious interest in finding an explanation outside myself. What I can say is that my account is not unusual in the published record, and the record is specific enough to check.

Mephedrone (4-methylmethcathinone) spread through Britain from the summer of 2009, sold openly online, in head shops and at festivals, and was made a Class B drug on 16 April 2010. The British Crime Survey found last-year use of 1.4 percent among adults aged 16 to 59 in 2010-11, and 4.4 percent among 16 to 24 year olds [31]. The European Monitoring Centre for Drugs and Drug Addiction (EMCDDA), in its 2010 risk assessment of mephedrone, reported a large online survey of dance-scene respondents in which "most participants had become aware of mephedrone during 2009 through coverage in the mass media and on the Internet," and recorded that some users said they first bought and used mephedrone after reading reports about it in the popular press [32]. The EMCDDA labelled that material anecdotal, and the survey was self-selected. Both qualifications stand. So does the direction of the finding, from the one dataset that asked users how they had first heard of the drug.

Forsyth, writing in the International Journal of Drug Policy in 2012, tracked news coverage against Google search behaviour and produced the closest thing this field has to a sequence [33]. Before the "Brighton death" of November 2009 there was little press coverage; after it, coverage and searches rose together, with search volume doubling. In March 2010, coverage of two teenage deaths in Scunthorpe spiked, and within days searches for "mephedrone" trebled, with the news increase preceding the search increase; searches for "mephedrone buy" rose further still, and automated advertisements for sellers appeared beside online news reports of the deaths. The Guardian reported in December 2009 that one consequence of that week's coverage was that many suppliers had sold out. Both anchor stories were false alarms. The Brighton death was found to be from natural causes, and it later emerged that the Scunthorpe teenagers had not taken mephedrone. The Advisory Council on the Misuse of Drugs reached similar ground in its 2010 cathinones report: the most popular related search was "buy mephedrone online," four of the top five search terms combined "buy" with "mephedrone," retailers reported increased sales after media coverage, and the Council recorded its concern that media attention may have brought these drugs to a wider demographic sooner than would otherwise have happened [34]. Alexandrescu's discourse analysis of roughly 110 tabloid items described a moral-epidemic frame built on metaphors of disease, moral downfall and invasion, in coverage that publicised at the same time the properties users valued: purity, availability, stimulant effects, low price and legality [35]. Measham and colleagues summarised the press role as acting "akin to an advertising agency with a contradictory brief" [36].

The confounder has to be stated with the same force. Measham and colleagues also showed that mephedrone's rise coincided with a collapse in the availability and purity of ecstasy and cocaine in 2009, which left experienced users looking for a legal and reliable substitute [36]. Qualitative studies in Northern Ireland and Ireland found initiation tracked price, availability and the poor quality of other stimulants more than it tracked headlines [37,38]. Search data, as the Home Office evidence review noted in 2014, show interest in a substance, not purchases or use [39]. No study demonstrates that media coverage caused population-level mephedrone use. The defensible claim is the one the record supports: coverage demonstrably raised awareness and purchase-directed curiosity, in a market where the product was legal, cheap and one search away, and users themselves named the media as their route in. My own history is one instance of that pattern. It is the pattern, not my instance of it, that carries weight for policy.

Awareness also did work on the other side of the clinic door, and fairness requires recording it. Kolliakou and colleagues, analysing a large London mental health record in 2016, found that rises in searches and page views for mephedrone preceded a peak in references to it in clinical notes, with spikes in all sources around classification, which they read as news and policy events driving clinical attention as well as public interest [61]. A clinician cannot ask about a drug he has never heard of, and some of the same coverage that recruited users also taught emergency departments what was arriving. That dual effect is exactly why the argument of this article is about how publicity is built, and what product detail it carries, rather than whether authorities should speak at all.

Britain then ran the larger experiment. Post-ban surveys showed supply moving from websites to street dealers, with access falling and past-month use declining in the dance-scene samples [32,36]. The Psychoactive Substances Act 2016 imposed a blanket ban on novel psychoactive substances. The Home Office review of the Act, published in 2018, found a considerable reduction in use in the general adult population and the end of open head-shop retail, and stated that there was insufficient evidence to establish whether displacement to other drugs had occurred in that population [40]. In the groups with the least protection, the picture was worse. Ralphs and colleagues reported in 2021 that the Act was associated with synthetic cannabinoids moving into illicit street markets, with more variable content and potency and greater harm to homeless users [41]. Deen and colleagues counted deaths with novel psychoactive substances detected at post-mortem at 91 in the three years before the Act and 202 in the three years after, driven by synthetic cannabinoids, in a voluntary reporting system where better detection may inflate the later figure [42]. A ban can shrink a market and still concentrate its harms on the people least able to absorb them. Both halves of that sentence belong in any discussion of 7-OH scheduling.

The case for acting, stated fairly

Nothing in the sections above argues that regulators should have left concentrated 7-OH alone, and the record contains a real case for action that this article accepts. Poison centre reports rose steeply in the

years when high-potency products spread, reaching 3,434 reports in 2025 [6]. Clinicians have published case reports of dependence and opioid-like withdrawal tied to smoke-shop 7-OH tablets, in people who bought a product sold in gummies, shots and tablets with no standardised dose and, in many places, no age restriction at the till [5]. A market like that is not a neutral backdrop. It advertises too, with packaging and placement that imply a supplement rather than an opioid, and official silence would leave that implication standing. Warning letters forced products off shelves and changed what could be marketed. Seizures removed supply. Public announcements also carry information that some current users will act on in the intended direction, and they alert clinicians, poison centres and parents to a presentation they may not have seen. The FDA's approach in 2025 was, by historical standards, a calibrated one: it drew a line between the concentrated product and the leaf, stated a toxicological basis for the line, and aimed enforcement at manufacturers and distributors. State policy shows the range on either side of that calibration, from full bans on all kratom products in six states and the District of Columbia to age limits and concentration caps in Colorado and Oklahoma [12]. Where a middle option exists in policy, the proportionality question deserves the same care in communication: the louder and broader the warning, the wider the audience of the previously unaware who receive it.

What this means for 7-OH, stated as hypothesis

No study has yet measured whether publicity about 7-OH has increased curiosity, experimentation or substitution to other opioids. Anyone claiming that finding exists is ahead of the literature. What exists is a mechanism with meta-analytic support, a set of campaign evaluations showing null and iatrogenic results under identifiable conditions, a British case in which death coverage preceded spikes in purchase-directed searching for a legal product, and a well-documented general pattern in which restricting one opioid supply channel can shift some users to another. On that last point, Evans and colleagues found that the 2010 abuse-deterrent reformulation of OxyContin was followed by substitution toward heroin, with no reduction in combined opioid and heroin mortality; a later re-analysis disputes the attribution, and the dispute itself is a reason to monitor substitution rather than assume it away in either direction [43].

Applied to 7-OH, the risk factors from the earlier sections are easy to list. The product was, until recently, obscure, and obscurity is the condition under which publicity does the most introducing. It has been sold in ordinary retail settings, which means a curious person does not need a dealer to act on curiosity. Announcements have, of necessity, named the product, described its effects in terms of opioid potency, and been reported alongside where it is sold. And the audience most at risk of reactance, adolescents and young adults with high sensation seeking, is the audience least likely to be persuaded by a federal warning and the most likely to hear a prohibition as a product endorsement from the one source whose disapproval carries social value. Against that, the FDA's own framing offers a partial model: it separated the concentrated product from the leaf in public, gave a toxicological reason for the line it drew, and

aimed its enforcement language at sellers. Communication that distinguishes products, explains the reason for a rule, and avoids repeating price, availability and effect detail is closer to what the evidence supports than a general alarm about a new drug sweeping the country.

The practical proposals that follow from the literature are about design and measurement rather than volume. Prevention messages should be pretested for reactance in the audience they target, including measures of anger and counterarguing, before they are scaled. Announcements of enforcement action should be audited for the product information they carry, the way a drug label is audited, and stripped of sourcing detail that serves curiosity rather than safety. Campaigns should be evaluated against behavioural outcomes with a standing plan to detect iatrogenic effects, as Hornik's team did, rather than against reach and recall. And the effect of the announcement should be monitored with the same instruments that exposed the mephedrone sequence: search behaviour, poison centre calls and treatment presentations, read together, in the weeks after each major publicity event.

Addiction as a disease, without a life sentence of identity

The author holds the position, common in family medicine, that addiction is a disease. The anchor study is McLellan and colleagues in JAMA in 2000, which set drug dependence beside type 1 diabetes, hypertension and asthma and found relapse rates of 40 to 60 percent for drug dependence, 30 to 50 percent for type 1 diabetes and 50 to 70 percent for hypertension and asthma, with comparable problems of adherence across all four, concluding that drug dependence should be insured, treated and evaluated like other chronic illnesses [44]. Volkow, Koob and McLellan set out the brain disease model in the New England Journal of Medicine in 2016, describing desensitised reward circuits, strengthened conditioned responses and weakened executive control [45]. The American Society of Addiction Medicine defines addiction as "a treatable, chronic medical disease involving complex interactions among brain circuits, genetics, the environment, and an individual's life experiences" [46]. The model has serious critics. Hall, Carter and Forlini argued in The Lancet Psychiatry in 2015 that the brain disease account is stronger in its advocacy than in its evidence, that it has not delivered the treatment advances promised for it, and that many people recover without treatment [47]. A commentary can hold the clinical value of the disease model, which is that it moves addiction out of the category of moral failure and into the category of conditions a health system is responsible for, while accepting that the neuroscience is incomplete.

The fracture analogy the author uses with families runs as follows. A person who breaks a bone may face surgery, a period in the sick role, a period of recovery and, in a minority of cases, lasting problems. Nobody suggests that the person's identity is now "fracture patient" for life, and nobody builds a philosophy on the claim that a refracture is a necessary part of having healed. Diabetes and heart disease sit at the other end: lifelong conditions, managed day to day, with exacerbations met by a care plan rather than by shame. Addiction care often takes the chronicity of the second group and adds a permanent

identity that neither group carries. The person is told, and tells himself, that he is an addict, always, and that relapse is part of recovery. There is a protective intention inside that teaching, since overconfidence kills people in early recovery, and fellowships that use this language have evidence behind their programmes for alcohol use disorder [48]. But the identity claim deserves the scrutiny any other clinical claim would get, because it has measurable effects.

The most direct evidence is Kelly, Greene and Abry's national randomised vignette study, published in Addiction in 2021. Describing addiction as a "chronically relapsing brain disease" produced the lowest blame of any framing they tested, which is the disease model doing its intended work. The same description also reduced prognostic optimism, and increased perceived need for continuing care and perceived dangerousness, compared with describing addiction as a problem [49]. One framing bought freedom from blame at the price of hope. Kelly's earlier work with Westerhoff found a related effect among clinicians themselves: the same person described as a "substance abuser" was judged more personally culpable and more deserving of punishment than the same person described as "having a substance use disorder" [50]. The honest summary of this literature is that language studies measure attitudes and judgements, not relapse or retention, so claims for clinical benefit from any particular vocabulary outrun the data. What the data do support is narrower and still important. Labels shift blame, punishment and expectations of recovery, in the public and in professionals. A framing that lowers blame while lowering expected recovery is a trade, and trades should be made deliberately, patient by patient, rather than installed as a permanent identity by default. NIDA's guidance on words points in the same direction and adds a point of respect: use person-first language, and let individuals choose how they are described, since some people in recovery claim the word "addict" for themselves inside fellowships and find strength in it [51]. The objection in this article is to the label as a life sentence handed down by the system, not to a word a person picks up voluntarily as a tool.

Population data give the objection a denominator. Kelly and colleagues' National Recovery Study, published in 2017, estimated that 9.1 percent of United States adults, about 22.35 million people, reported that they used to have a problem with alcohol or drugs and no longer do, and roughly half of them reported resolving the problem without formal treatment services [62]. Whatever route those people took, the outcome they report is resolution, in the past tense. A framing of addiction that cannot describe 22 million people accurately is a framing with a defect. The chronic disease model survives that observation easily, since diabetes care also contains people whose condition remits and stays remitted. What does not survive it is the stronger claim, heard in treatment rooms and repeated in families, that resolution is never really resolution and that the correct self-description for the rest of a life is the name of the disease at its worst.

Care built around the person

If campaigns should be designed against reactance, care should be designed around the individual. The research base for the components is uneven, and this section grades it as it goes, because a commentary that demands evidence from regulators should demand it from itself.

Motivational interviewing, which works by drawing out a person's own reasons for change rather than arguing them into it, has small effects in the meta-analytic literature. Lundahl and colleagues pooled 119 studies in 2010 and found a small durable effect against weak comparison conditions (g = 0.28) and no significant advantage over specific active treatments (g = 0.09) [52]. Cognitive behavioural approaches show a similar profile, with a small overall effect across 53 trials in Magill and Ray's 2009 synthesis [53]. These are modest numbers. They support psychological services as a standing part of addiction care, delivered by people trained in them, without supporting the claim that counselling alone resolves a condition with the relapse profile McLellan documented.

Harm reduction occupies a specific place in this argument, and the author's position is the evidence position: measures that reduce the harm of ongoing use, paired with a route toward stopping, with a goal and a timescale the person has agreed. The European Monitoring Centre's harm reduction monograph records sufficient evidence that opioid substitution treatment reduces HIV transmission, and no strong evidence that substitution treatment, needle and syringe programmes or consumption rooms, when well managed, increase harms or encourage wider use [54]. Three decades of work on syringe services programmes, summarised by NIDA in 2024, found no increase in drug use or crime, with linkage into treatment [55]. Harm reduction without an agreed direction becomes maintenance of a situation nobody chose. Harm reduction with a coercive deadline becomes something else again, and the literature warns against it directly: studies of discontinuation of opioid agonist treatment report very high rates of return to use after imposed or rapid tapers, and cohort data associate slower, individualised tapering and longer treatment duration with lower overdose risk [56]. A finite goal is defensible when it is the patient's goal, set the way a quit date is set in smoking cessation, reviewed against how the person is doing, and revised without penalty when circumstances change. The smoking literature supports the dated target itself: Lindson and colleagues' 2019 Cochrane review found reduction-to-quit and abrupt quitting produced similar long-term abstinence (relative risk 1.01), with most reduction programmes still working toward a quit day [57]. What transfers is the structure, a named day and a plan that converges on it, held by the person whose day it is.

Care also sits inside a circle wider than a clinic. Families, employers, faith communities and peer groups shape whether a person in early recovery has somewhere to be, someone expecting him, and a role that is not defined by his worst period. The evidence for mutual-help involvement is strongest for alcohol: Kelly, Humphreys and Ferri's 2020 Cochrane review of 27 studies and 10,565 participants found

manualised Alcoholics Anonymous and twelve-step facilitation produced higher rates of continuous abstinence at 12 months than other established treatments (relative risk 1.21), with cost savings in most economic analyses [48]. That finding is about alcohol and about a specific, structured form of community involvement, and it should not be stretched to cover every drug or every programme. For faith-based interventions as a category, Hai and colleagues' 2019 meta-analysis of 20 randomised trials found an effect against inactive control that was moderate in size and statistically nonsignificant, and comparisons were dominated by twelve-step-oriented programmes, so efficacy claims for faith-based treatment in general are not currently supported [58]. What faith communities can supply, on engagement rather than efficacy grounds, is what congregations in fact supply to many people in recovery: belonging, routine, moral seriousness without clinical language, and a welcome that does not begin with a diagnosis. A whole-person model that a family physician would recognise puts these elements around the clinical core: the named clinician who knows the patient, psychological support, physical health treated in the same plan rather than in a separate queue, family involvement with the patient's consent, and community and faith stakeholders where the patient wants them. It also names an end to the intensive phase. A person who reaches sustained remission should be able to graduate from a treatment identity the way a person with controlled diabetes steps down to annual review, with the door open and no ceremony of failure if he returns.

Limitations

This is a commentary, and its central policy claim is a risk claim rather than a demonstrated effect. No study shows that publicity about 7-OH has increased its use, and the mephedrone evidence is ecological, self-reported or drawn from self-selected samples, confounded by the collapse of the ecstasy and cocaine markets in the same years. The campaign literature contains a genuine counter-example in the SENTAR work, showing that mass messaging aimed at high-risk adolescents can reduce use when designed for its audience. Effect sizes across the prevention literature are small in both directions, which means the argument here supports careful design and monitoring, not silence. A regulator may reasonably judge that the toxicological case for scheduling concentrates is strong enough to accept a publicity risk; this article argues the risk should be measured and managed rather than treated as zero. The author's lived experience is disclosed as part of his motivation and is a source of bias as well as insight; readers should weight the citations, which stand or fall without it.

Conclusion

A warning is an intervention. It has a mechanism, a target population, a dose and adverse effects, and for little-known drugs one of its effects is introduction. The record from DARE, from the national anti-drug campaign, from tobacco and from mephedrone is consistent enough to act on: messages that lecture,

threaten and repeat the product's name, price and availability can move part of their audience toward the behaviour they oppose, and messages that respect autonomy and redirect distrust toward the seller can move an audience the other way. The current response to concentrated 7-OH may prove to be the right regulatory call. It will be a better one if its communication is designed and tested like the intervention it is, if enforcement announcements stop carrying a product catalogue inside a warning, and if the care offered to the people already using these products treats them as patients with a chronic disease and a future, with goals they set, support that wraps around the whole of a life, and an exit from the identity of addict that they are allowed, in time, to walk through.

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